

Segmented Encryption: A Quality and Safety Supervisory Model for Herbal Medicine Based on Blockchain Technology

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Abstract—The quality of herbal medicine has an important impact on human health. In this paper, we proposed a blockchain-based herbal quality and safety supervisory model for the current frequent herbal counterfeiting phenomenon. We manage the production, processing, and trading processes of herbal medicines by exploiting the blockchain's immutable and traceable properties. We proposed a segmented encryption method for information to encrypt the private information of enterprises. We use shared cloud storage to reduce waste of local storage space, and we proposed a verifiable random chain cutting mechanism based on the verifiable random function to handle the redundant blocks of the chain. Our article addressed the problem of herbal source falsification. Automated recording of key factors such as soil and temperature that affect the quality of herbal medicines is done from the seedling stage. Our herbal quality and safety supervisory model used blockchain technology to increase control over the production and distribution of herbal products, reduce herbal counterfeiting, and improve the efficiency of the system.

Index Terms—herbal medicine, blockchain, the Internet of things, production supervision, traceability.

I. INTRODUCTION

Herbs are not only used for culinary purposes but are also an ancient source of spices and medicines. As socio-economic and ideological levels increase, herbal medicine for health care is more and more accepted. Numerous scientific studies have found that medicinal herbs and some of the ingredients they contain have preventive and inhibitory effects on certain diseases. Studies have shown that natural phenolic compounds from herbal medicine play an important role in cancer prevention and treatment [1], [2]. Some herbs such as *scutellaria baicalensis* Georgi and its identified components which have been shown to inhibit infectivity and replication of HIV [3]. In some Asian and African countries, medicinal herbs in Lamiaceae family have is commonly used to treat hypertension [4]. Many reports have shown the positive effects of herbal extracts as an antiviral agent in animal feed or for prevention and treatment. The use of herbs may also reduce the development of drug resistance and may modulate the immune system to prevent virus-related diseases [5].

During the COVID-19 outbreak in early 2020, herbal medicine played an important role in the antiviral process. Of the more than 80,000 cases diagnosed in China, more than 70,000 were treated with herbal medicine, accounting for 91% of the total. More than 90 percent of patients in Hubei Province have received herbal treatment to various degrees. The advantages of Chinese herbal medicine in preventing and controlling epidemics have been highlighted, and a wide range of effective prescriptions and herbal medicines have been widely used.

As a special commodity for the prevention, diagnosis, and treatment of diseases, herbal medicine are closely related to people's immediate interests and social harmony and stability. However, in the current market environment, it is difficult to ensure the quality of medicinal herbs due to some illegal business, processing and adulteration problems.

In February 2016, six batches of Saint John's Wort products manufactured in 2013 and sold in the UK were recalled due to higher levels of pyrrolizidine alkaloids (PAs) than recommended by the European Medicines Agency [6]. Authorities had difficulty determining the source of the adulterated material and where the incriminated batches ended.

A study by Booker et al. [7] of the quality of herbal products on the market found that some samples contained undeclared edible dyes. The concentrations of active ingredients in the samples were much lower than those of high-quality herbs.

Herbal products are susceptible to heavy metal contamination, especially in rapidly developing countries such as China and India. Pollution from heavy industry is a particular problem, as some plants naturally accumulate heavy metals, which severely affects the quality of herbal medicines [8]. We need to regulate the environment in which herbs are cultivated.

In order to ensure the quality of herbal medicine, it is necessary to understand the factors that affect its quality and to propose new solutions. What technology can solve these problems with herbs? The emergence of blockchain technology quickly attracted people's attention. Driven by both policy and market demand, blockchain technology is rapidly expanding in the medical field. Blockchain technology can be used to track the whole process of herbal medicine

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production, processing and sales, which perfectly fits with the supply chain system and increases the control ability of herbal medicine products. Each block on the blockchain has a timestamp to form a chain. From the production, processing, transportation, and sale of herbs, the chronological transactions form a complete record of the whole process.

The immutable and decentralized characteristics of the blockchain can increase credit endorsement. The nodes on the chain are able to supervise each other. Moreover, due to too much information uploaded to the chain, it also increases the risk and difficulty of falsification. Blockchain technology can also reduce supply chain costs. It establishes connections to various links in the supply chain to optimize the supply chain of herbal products, which can reduce unnecessary double-validation and thus improve efficiency.

Using blockchain technology can better clarify responsibilities and strengthen credit registries. Once there is a problem with which link, it is easy to locate the responsible unit and the corresponding responsible person after the incident. It will be much more difficult for fakes to pass off as good products.

This paper will introduce the application of blockchain technology in the supervisory process of herbal production, transportation, and trading.

The rest of the paper is organized as follows. Section II introduces the related work. Section III describes the specific design of some practical problems. In Section IV the systematic implementation and evaluation of the scheme are expounded. Lastly, our conclusion is presented in Section V.

II. RELATED WORK

A. Blockchain Background

Blockchain technology was born from Bitcoin, which uses underlying technologies such as cryptography distributed storage and consensus algorithms. These technologies can be applied not only to the digital currency but also to other areas. After the comprehensive refinement of these technologies, the concept of blockchain emerged, which is not a single technology, but a combination of various technologies. Blockchain is a distributed, immutable, transparent, secure, and auditable ledger that records all executed transactions and is shared among all participants. It is also an open global infrastructure that allows companies and individuals to make transactions to reduce intermediaries, thereby reducing transaction costs and time [9].

Fig. 1 is an example of a blockchain that consists of a continuous block sequence. The data in a blockchain is stored in a block and connected one by one by hashing, thus creating a chain, so it's called the blockchain. Each transaction is agreed upon by the participants. Information about each completed transaction in the blockchain will be shared with all nodes. Once a transaction is recorded in the ledger, it cannot be erased [10]. Each block contains the hash value of the previous block header (except the Genesis block). All transactions will be stored in the blockchain. If a block is to be tampered with, the parent hash contained in the next block will not match the

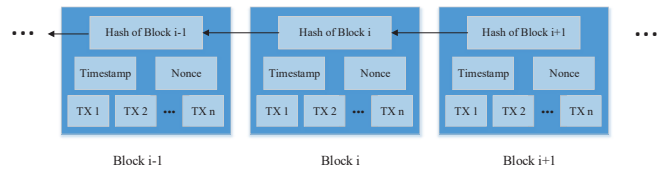


Fig. 1. An example of blockchain which consists of a continuous sequence of blocks.

value tampered with, and the tampering will be easily detected.

B. Blockchain of herbal supervisory

At present, there are few studies on the use of blockchain technology in the quality supervision of herbal medicine. Some studies have focused on the sharing of electronic medical records established after herbal medicine consultation, while some used the Internet of Things coupled with RFID technology to trace herbs back to their origins [11].

Juan F. Galvez et al. [12] examined the potential of blockchain technology to ensure traceability and authenticity in the food supply chain. It can be considered as a truly innovative and relevant approach to ensure the quality of data acquisition and management of analytical processes. Jen-Hung Tseng et al. [13] used blockchain as the basis for drug data flow to create transparent drug transaction data.

By analyzing cases of herbal counterfeiting, Michael Heinrich et al. [8] argued that transportation, storage and processing are key areas where blockchain management can help minimize risk.

Most of the current management solutions based on blockchain are focused on food and pharmaceuticals, for which we propose a new supervisory approach for the herbal. We hope to better regulate the quality of herbal medicines from the source.

III. SPECIFIC DESIGN

A. Segmented encryption of information

In the whole medicinal herbs quality and safety supervisory model, the participating nodes include planting base, processing manufacturers, transportation companies, distributors, retailers and consumers, etc. Due to the large number of entities involved and the multiple sources of heterogeneous data, not all information can be shared. The system should also pay attention to the protection of data privacy while safeguarding the information from being tampered with. In this paper, a solution of information segmented encryption is proposed to encrypt the private data of an enterprise. Only authorized nodes can see the private data of the enterprise, while other nodes without permission can only see the hash value of the private data, so as to ensure the data privacy of the enterprise [13].

The cryptographic algorithms utilized in the information segmentation encryption model are symmetric cryptographic algorithm AES and asymmetric cryptographic algorithm RSA.

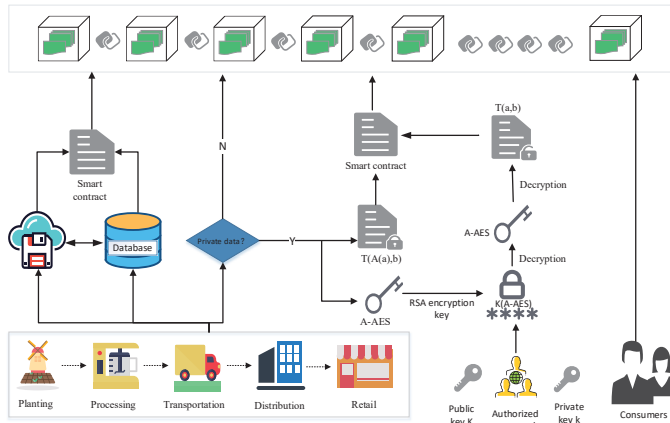


Fig. 2. Privacy data protection for herbal production and supply.

The Symmetric cryptographic algorithm means that the same secret key is used for encryption and decryption. Whereas the asymmetric encryption algorithm requires public and private keys. If the data is encrypted with the public key, it can only be decrypted with the corresponding private key. As shown in Fig. 2, when enterprise node A initiates a transaction $T(a, b)$, where a is private information such as enterprise production process data that needs to be kept confidential, and b is enterprise information and herbal information that can be made public. Enterprise node A uses AES symmetric encryption algorithm to generate its own key A-AES. It then uses the key A-AES to encrypt the private data into $A(a)$, which becomes a semi-public transaction $T(A(a), b)$ and sends it to the blockchain network for consensus. We define the public key of the authorization node or supervisory node that can view the private data as K . The enterprise node A uses the public key K to generate $K(A-AES)$ with RSA asymmetric encryption of the enterprise's A-AES key and sends it to the corresponding authorized nodes.

Authorized nodes and supervisory nodes can use their own private key k to decrypt $K(A-AES)$, get the key A-AES of enterprise A and use the key A-AES to decrypt the encrypted semi-public transaction $T(A(a), b)$ to get $T(a, b)$, which can get the information they need. Here the private data can be authorized and viewed by multiple nodes. Different authorized nodes only need to invoke the smart contract multiple times to view the data, and the data permissions do not affect each other.

B. Shared cloud storage

The herbal medicine base cultivation chain includes seedlings, irrigation water, soil, air, and other key factors that affect the quality of herbal medicine. A great deal of practical experience has shown that the better seedling variety is selected, the better quality of the herbs. The quality of herbs can vary greatly depending on the environment and climatic conditions in which they are grown. By recording these growth conditions and other information on the chain,

a traceable record of the entire herbal production process can be made, which facilitates finding the causes that affect the quality of the batch of herbs. Since the data that exists on the chain is typically structured, it aids in data analysis. If this information is studied, it is more able to find out the growth patterns of the herbs from which better and higher quality herbs can be cultivated. We want to take advantage of the immutable nature of the blockchain to document these critical factors and enable nodes in the chain to share this information. In this herbal medicine quality and safety supervisory model, the main focus is on the traceability of the production records of herbal medicines. Fig. 3 shows a flowchart of information collection and uploading of herbal medicine production. For example, the name of herbal medicine, the person in charge of the area and the brand of fertilizer can be input into the supervisory system. Some information such as temperature, humidity, image information and geographical location can be automatically collected by the system through sensors. Each sensor has a wireless data transmitting module and data storage module, which can send the collected data in real-time.

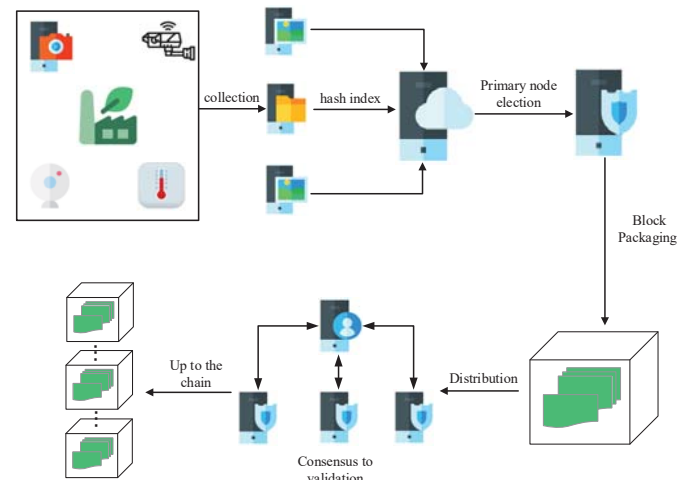


Fig. 3. The process of uploading herbal production information to the blockchain.

Since images and other information occupy large storage space, we use cloud storage to process such information. When there is a need for a query, we use key-value pair indexing to get the required information. A hash function is a method of creating a small numerical "fingerprint" from any kind of data. The hash function compresses the message or data into a digest, fixing the format of the data so that the amount of data is small. The function scrambles the data and mixes it to recreate a fingerprint called the hash value. The hash value is usually represented by a short string of random letters and numbers.

After the data in the system is collected in real-time, the SHA256 hash function is used to hash the data and image information to get a message digest. For messages of any length, SHA256 generates a 256-bit long hash, called a message digest. This digest is equivalent to an array of 32 bytes in length, usually represented by a hexadecimal string

of length 64. Fig. 4 illustrates the process of producing a hash index for herbal production information before cloud storage. The system hashes the image information and the text information of the production record separately by using a SHA256 hash and finally gets a root hash to represent the index value of that production record. Nodes on the blockchain can use this index value to find the information they want.

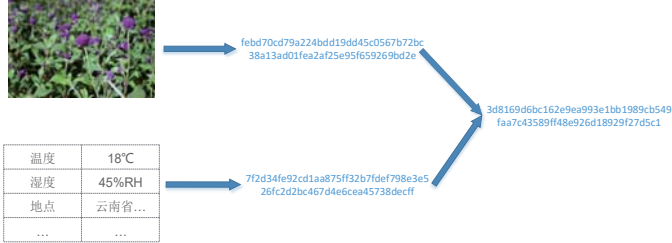


Fig. 4. Hash index of production information.

C. Verifiable random chain cutting mechanism based on the VRF

The data on the blockchain grows over time. After a period of time, some herbal data on the chain may become obsolete, generating a large number of redundant blocks. To avoid spatial redundancy, we propose a verifiable random chain cutting mechanism based on the VRF (verifiable random function, VRF) to avoid infinite expansion of data by removing unwanted blocks. This will keep the entire blockchain data volume within a certain range.

Verifiable random function (VRF) is proposed by Silvio Micali et al [14] in 1999. VRF is essentially a kind of pseudo-random function with verification function. It can be regarded as a Random Oracle (RO), that is, a random number output can be obtained through any input. The verifiable random function has a non-interactive zero-knowledge proof over the random predictor, which can be used to verify the correctness of the output of the random number, indicating that the random number is indeed born by someone.

There are three types of node identities in this cutting mechanism, the supervisory node, the cutting node and the slave node. As shown in Fig. 5, the nodes selected by the VRF algorithm are called cutting nodes. The cutting node starts from the initial block of the chain to find information about the expired block. They find the nearest expired block and distribute the cutting information such as block height over the network. After the slave nodes received the information from the cutting node, the public key of the cutting nodes are first used for signature authentication. Secondly, the message is hashed (message digest to reduce the size of the information's transport across the network) and signed and sent to the supervisory node. When the supervisory node received a cutting message sent by $2f$ [15] (The maximum number of Byzantine fault nodes occurring simultaneously in this system is f) nodes and all signatures are verified, the chain cutting step can be performed and broadcast to other nodes across the network.

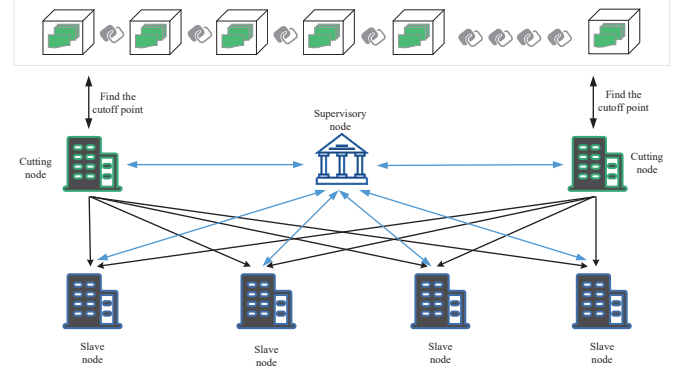


Fig. 5. The process of chain cutting.

The complete process can be described simply as follows:

1) *Step 1:* The selected cutting node i has its public key PK_i and private key SK_i , and generates two values, *Result* and *Proof*, with the random number given by the system *info*:

$$\text{Result} = \text{VRF_Hash}(SK_i, \text{info}) \quad (1)$$

$$\text{Proof} = \text{VRF_Proof}(SK_i, \text{info}) \quad (2)$$

2) *Step 2:* The cutting node hands over the *Result* and *Proof* to the slave and supervisory nodes. The slave nodes and the supervisory node confirm whether node i is a cutting node by validating the value of the *Result* after receipt. When the identity of node i is confirmed, the identity of node i is confirmed through the public key of node i and *Proof* to ensure that the information has not been falsified.

$$\text{Result} = \text{VRF_P2H}(\text{Proof}) \quad (3)$$

$$\text{True/False} = \text{VRF_Verify}(PK_i, \text{info}, \text{Proof}) \quad (4)$$

The advantage of VRF algorithm is that in the whole operation process, the cutting node never shows its private key, while the verifier can deduce whether *info* and *Result* match accordingly.

IV. IMPLEMENTATION AND EVALUATION

A. System implementation

The blockchain-based herbal medicine quality and safety supervisory model can monitor and track the data generated from each stage of herbal medicine production to distribution in real-time and process them accordingly. This article is based on the Fabric framework introduced by IBM for system development, using Docker containers and shell scripts to build and configure the blockchain network, using the go language to write relevant smart contracts and backend servers, and using Vue as the front-end framework for the user to provide a corresponding front-end web page for operation.

Records have a unique traceability barcode or QR code on the package of products in the blockchain-based herbal quality

and safety supervisory platform. When consumers use their mobile phones to scan the identification code on the package, they can quickly find out the information about the product's record on the blockchain network. Fig. 6 shows an example of the supervisory platform. The consumer can get the production and traceability information of the product by scanning the QR code or barcode on the product package through the applet. From Fig. 6, we can see the details of the herbal medicine production and the chain data. Consumers can view the corresponding production, processing, and transportation information of the purchased products through the applet. The system makes use of the immutable nature of the blockchain network to ensure the authenticity and reliability of the data on the chain viewed by consumers.

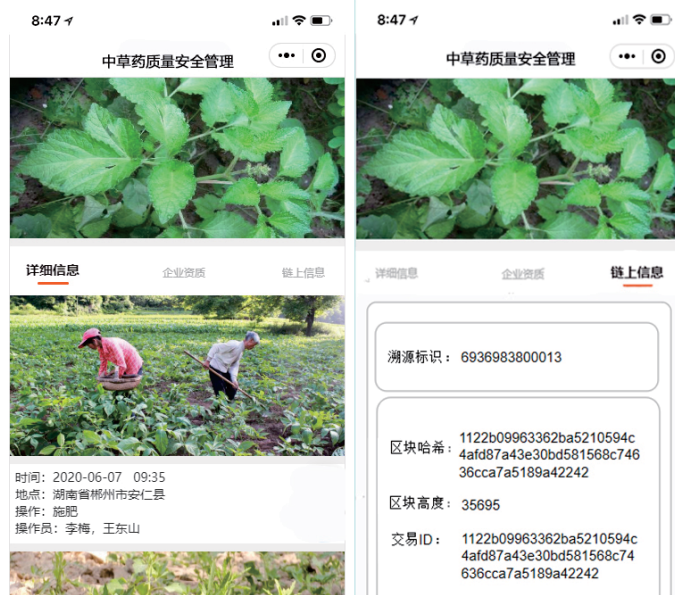


Fig. 6. Consumer Interface.

Fig. 7 shows the interface of an enterprise user using the supervisory platform. The enterprise can use this platform to propose transactions to cooperate with other entities. The transaction contract and partner information will be recorded on the chain to facilitate the review of supervisory authorities and to reduce corruption within the enterprise. Enterprises can only view public information on the chain and information that has been authorized to be viewed. They can only see the cryptographic hashes of unauthorized information.

Fig. 8 shows the interface where the supervisory node manages the users on the chain. The supervisory node determines whether the node can register and join the system based on the industry assessment and the real identity information submitted.



Fig. 7. Enterprise User Interface.



Fig. 8. Supervisory Node Interface.

B. System safety analysis

In this system, an unauthorized node that wants to obtain the encrypted information can choose to violently crack the AES algorithm to get the encrypted secret message, or choose to crack the authorized node asymmetric encryption RSA algorithm to get the corresponding private key k .

Confusion is a criterion for testing the security of an encryption algorithm, where higher confusion means better encryption, making it impossible for an adversary to infer its key even if he acquires some statistical characteristics about the secret message. We measure the confusion of AES symmetric cryptographic algorithm by the secret message change rate P . $P = m/n * 100\%$, where m is the number of changed bits of the secret message and n is the total number of bits of the secret message. In this experiment, we control the plaintext to remain unchanged and change one secret key bit for ten experiments, and the changes caused are shown in Fig. 9. It is verified that the AES encryption algorithm has a high confusion with an average value of 0.831 for the rate of change of the secret text.

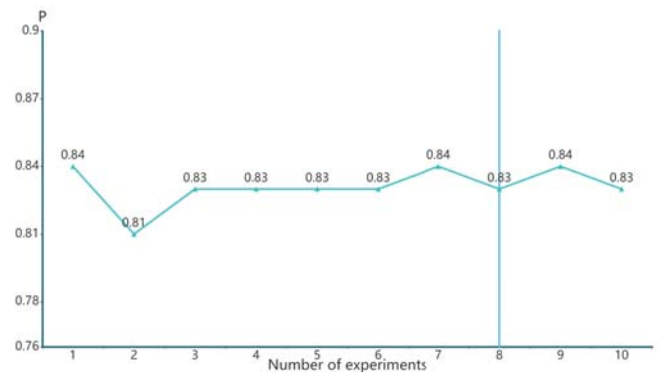


Fig. 9. Confusion testing.

Obtaining the private key k of an authorized node requires breaking the RSA encryption algorithm, which is one of the most widely used asymmetric encryption algorithms. Its security relies on the difficulty of the factorization of large integers. Therefore theoretically our model is safe and reliable to store the data on the blockchain network using this method.

C. Performance testing

We use Hyperledger Caliper, a blockchain performance testing framework that allows users to define a test set, perform a performance test against their own blockchain network, obtain a series of test results, and generate test reports. We start the client child process to test and return statistics through a predefined benchmark configuration file.

Caliper provides a set of interfaces for the application to interact with the back-end blockchain system. We conducted seven rounds of testing, testing the throughput and latency of 100 transactions at different sending rates, and as seen in Fig. 10, the average latency of each transaction is less than 1s, which meets the needs of a real supply chain.

Succ	Fail	Send Rate(TPS)	Max Latency(s)	Min Latency(s)	Avg Latency(s)	Throughput(TPS)
100	0	15	1.13	0.48	0.85	10.2
100	0	19.2	1.19	0.4	0.68	17.5
100	0	19.8	1.14	0.43	0.7	12.1
100	0	20.6	1.4	0.41	0.85	19
100	0	21.7	1.41	0.41	0.92	12.2
100	0	51.6	0.18	0.04	0.09	50.2
100	0	66.7	0.17	0.02	0.06	66

Fig. 10. Results of performance testing.

V. CONCLUSION

In this paper, we design a model for herbal medicine quality and safety supervision by using blockchain technology. This model targets the problem of the source counterfeiting of herbal medicines relative to other traceability models. Sensors are used from the seedling stage to automatically record key factors such as soil and temperature that affect herbal medicine quality. By leveraging the decentralized, secure, and traceable features of blockchain, the credibility of the endorsement of the entire herbal supply chain is improved to reduce the risk of counterfeiting. This system protects the privacy and security of an enterprise's private data by using segmented encryption of information. We used shared cloud storage to solve the problem of local storage of large files. We proposed the verifiable random chain cutting mechanism based on the VRF to handle the redundant blocks. Finally, we showed our system and analyze the security and performance of the system, which meets the demand for quality and safe supply of actual herbal medicine and can integrate resources to improve the collaboration efficiency of the whole supply chain.

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